Final Project Proposal: Group 09

Group Members

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Description of the Problem

College students aged 18–21 are under increasing stress from multiple sources: academic workload, career pressure, social expectations, and health challenges. Chronic stress impacts not only academic success but also mental and physical well-being. Identifying the underlying causes of stress and predicting its type (Eustress, Distress, or No Stress) can help design interventions such as counseling, peer mentoring, and academic adjustments.

This project aims to analyze student stress survey data to identify key stressors, classify students into stress categories, and provide actionable insights for universities to promote student well-being.

Description of the Dataset

The dataset "Student Stress Monitoring Datasets" (Kaggle, 2022) was collected via a nationwide Google Form survey of 843 anonymized college students aged 18–21. It contains around 20 core features grouped into five categories:

Psychological Factors: anxiety_level, self_esteem, mental_health_history, depression
Physiological Factors: headache, blood_pressure, sleep_quality, breathing_problem
Environmental Factors: noise_level, living_conditions, safety, basic_needs
Academic Factors: academic_performance, study_load, teacher_student_relationship, future_career_concerns
Social Factors: social_support, peer_pressure, extracurricular_activities, bullying

Additional Key Variables:

- Demographic: Gender (0 = Male, 1 = Female), Age (18–21)
- Stress Indicators: stress_experience, palpitations, anxiety, sadness, loneliness, irritability, concentration issues
- Academic & Environment Stressors: workload, competition, confidence, class attendance, conflicts with instructors, study environment
- Social & Relationship Factors: relationship stress, lack of leisure, family/peer pressures
- Target Variable: Primary stress type (Eustress, Distress, No Stress)

Research Objectives

1. Identify the most influential factors contributing to student stress.
2. Develop machine learning models to classify students into stress categories (Eustress, Distress, No Stress).
3. Explore correlations between psychological, physiological, social, academic, and environmental stressors.
4. Provide evidence-based recommendations for universities to improve student well-being programs.

Comments & Concerns

Survey responses may contain subjective bias; robust modeling techniques will be considered. Some features may have class imbalance; preprocessing and resampling techniques may be applied. Stress is a sensitive health-related issue; ethical considerations must be ensured in interpretation and reporting.

References

Student Stress Monitoring Datasets